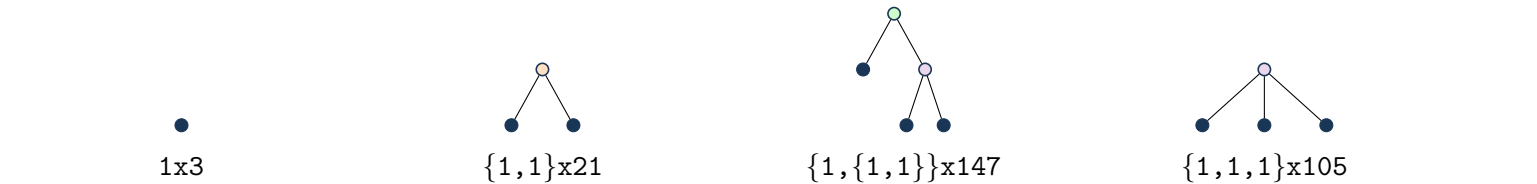


Definition and first terms

$16\,A(x) = 15 + 27\,x + A(x)^7$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+7*T^2+35*T^3+105*T^4+189*T^5+189*T^6+81*T^7$.
 $a(0), \dots, a(7) = 1, 3, 21, 399, 9135, 233709, 6400947, 183585897$.
 Kernel used throughout: $\rho(u) = u\,(1 - 7\,u^1 - 35\,u^2 - 105\,u^3 - 189\,u^4 - 189\,u^5 - 81\,u^6)$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.
 Typogeometric constructor codes and multiplicities: Δ_2 : 7, Δ_3 : 35, Δ_4 : 105, Δ_5 : 189, Δ_6 : 189, Δ_7 : 81.

Coefficient and contour extraction

$$\begin{aligned}
 \mathcal{K}_n &= \{(k_1, \dots, k_6) \in \mathbb{Z}_{\geq 0}^6 : \sum_{j=1}^6 j k_j = n - 1\}, \quad K = \sum_{j=1}^6 k_j, \\
 a(n) &= \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_6} 7^{k_1} 35^{k_2} 105^{k_3} 189^{k_4} 189^{k_5} 81^{k_6}. \\
 a(n) &= \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.
 \end{aligned}$$

Recurrence

$\sum_{r=0}^6 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-6 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 6. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$\begin{aligned}
 R(n, u) &= N(n, u) / \rho(u)^5, \quad \deg_u N = 36. \\
 \text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) &= \partial_u (R(n, u) H_n(u)).
 \end{aligned}$$

Matrix dimensions: 14×14 ; remainder 6×7 , rank 6, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.